

Math 534 HW 5 Question 4

Eric Graziosi

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'readr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2   3.5.2      v tibble    3.2.1
```

```
## v lubridate 1.9.4      v tidyr     1.3.1
```

```
## v purrr     1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.4.3
```

```
debt <- read_excel("C:/Users/ericg/Downloads/debt.xls")
```

```
growth <- read_excel("C:/Users/ericg/Downloads/growth.xlsx")
```

Step 1: Learn about the data collection and background

The debt dataset contains country-level debt as a percentage of GDP across years, while the growth dataset contains annual GDP growth for different countries.

Step 2: Load the data into R

```
debt
```

```
## # A tibble: 197 x 217
##   'DEBT (% of GDP)'   '1800' '1801' '1802' '1803' '1804' '1805' '1806' '1807'
##   <chr>              <chr>  <chr>  <chr>  <chr>  <chr>  <chr>  <chr>  <chr>
## 1 <NA>              <NA>   <NA>   <NA>   <NA>   <NA>   <NA>   <NA>   <NA>
## 2 Afghanistan      no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 3 Albania           no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 4 Algeria           no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 5 Angola            no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 6 Anguilla          no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 7 Antigua and Barbuda no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 8 Argentina         no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 9 Armenia           no data no data no da~ no da~ no da~ no da~ no da~ no da~
## 10 Australia        no data no data no da~ no da~ no da~ no da~ no da~ no da~
## # i 187 more rows
## # i 208 more variables: '1808' <chr>, '1809' <chr>, '1810' <chr>, '1811' <chr>,
## #   '1812' <chr>, '1813' <chr>, '1814' <chr>, '1815' <chr>, '1816' <chr>,
## #   '1817' <chr>, '1818' <chr>, '1819' <chr>, '1820' <chr>, '1821' <chr>,
## #   '1822' <chr>, '1823' <chr>, '1824' <chr>, '1825' <chr>, '1826' <chr>,
## #   '1827' <chr>, '1828' <chr>, '1829' <chr>, '1830' <chr>, '1831' <chr>,
## #   '1832' <chr>, '1833' <chr>, '1834' <chr>, '1835' <chr>, '1836' <chr>, ...
```

```
growth
```

```
## # A tibble: 265 x 70
##   'Country Name'      'Country Code' 'Indicator Name' 'Indicator Code' '1960.0'
##   <chr>              <chr>          <chr>           <chr>           <lgl>
## 1 Aruba              ABW            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 2 Africa Eastern and~ AFE            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 3 Afghanistan        AFG            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 4 Africa Western and~ AFW            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 5 Angola             AGO            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 6 Albania            ALB            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 7 Andorra            AND            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 8 Arab World         ARB            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 9 United Arab Emirat~ ARE            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 10 Argentina          ARG            GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## # i 255 more rows
## # i 65 more variables: '1961.0' <dbl>, '1962.0' <dbl>, '1963.0' <dbl>,
## #   '1964.0' <dbl>, '1965.0' <dbl>, '1966.0' <dbl>, '1967.0' <dbl>,
## #   '1968.0' <dbl>, '1969.0' <dbl>, '1970.0' <dbl>, '1971.0' <dbl>,
## #   '1972.0' <dbl>, '1973.0' <dbl>, '1974.0' <dbl>, '1975.0' <dbl>,
## #   '1976.0' <dbl>, '1977.0' <dbl>, '1978.0' <dbl>, '1979.0' <dbl>,
## #   '1980.0' <dbl>, '1981.0' <dbl>, '1982.0' <dbl>, '1983.0' <dbl>, ...
```

Step 3: Examine the data and create action items

```
sum(duplicated(debt))
```

```
## [1] 1
```

```
sum(duplicated(growth))
```

```
## [1] 0
```

```
names(debt)
```

```
## [1] "DEBT (% of GDP)" "1800" "1801" "1802"
## [5] "1803" "1804" "1805" "1806"
## [9] "1807" "1808" "1809" "1810"
## [13] "1811" "1812" "1813" "1814"
## [17] "1815" "1816" "1817" "1818"
## [21] "1819" "1820" "1821" "1822"
## [25] "1823" "1824" "1825" "1826"
## [29] "1827" "1828" "1829" "1830"
## [33] "1831" "1832" "1833" "1834"
## [37] "1835" "1836" "1837" "1838"
## [41] "1839" "1840" "1841" "1842"
## [45] "1843" "1844" "1845" "1846"
## [49] "1847" "1848" "1849" "1850"
## [53] "1851" "1852" "1853" "1854"
## [57] "1855" "1856" "1857" "1858"
## [61] "1859" "1860" "1861" "1862"
## [65] "1863" "1864" "1865" "1866"
## [69] "1867" "1868" "1869" "1870"
## [73] "1871" "1872" "1873" "1874"
## [77] "1875" "1876" "1877" "1878"
## [81] "1879" "1880" "1881" "1882"
## [85] "1883" "1884" "1885" "1886"
## [89] "1887" "1888" "1889" "1890"
## [93] "1891" "1892" "1893" "1894"
## [97] "1895" "1896" "1897" "1898"
## [101] "1899" "1900" "1901" "1902"
## [105] "1903" "1904" "1905" "1906"
## [109] "1907" "1908" "1909" "1910"
## [113] "1911" "1912" "1913" "1914"
## [117] "1915" "1916" "1917" "1918"
## [121] "1919" "1920" "1921" "1922"
## [125] "1923" "1924" "1925" "1926"
## [129] "1927" "1928" "1929" "1930"
## [133] "1931" "1932" "1933" "1934"
## [137] "1935" "1936" "1937" "1938"
## [141] "1939" "1940" "1941" "1942"
## [145] "1943" "1944" "1945" "1946"
## [149] "1947" "1948" "1949" "1950"
## [153] "1951" "1952" "1953" "1954"
## [157] "1955" "1956" "1957" "1958"
## [161] "1959" "1960" "1961" "1962"
## [165] "1963" "1964" "1965" "1966"
## [169] "1967" "1968" "1969" "1970"
## [173] "1971" "1972" "1973" "1974"
## [177] "1975" "1976" "1977" "1978"
## [181] "1979" "1980" "1981" "1982"
## [185] "1983" "1984" "1985" "1986"
```

```
## [189] "1987"      "1988"      "1989"      "1990"
## [193] "1991"      "1992"      "1993"      "1994"
## [197] "1995"      "1996"      "1997"      "1998"
## [201] "1999"      "2000"      "2001"      "2002"
## [205] "2003"      "2004"      "2005"      "2006"
## [209] "2007"      "2008"      "2009"      "2010"
## [213] "2011"      "2012"      "2013"      "2014"
## [217] "2015"
```

```
names(growth)
```

```
## [1] "Country Name" "Country Code" "Indicator Name" "Indicator Code"
## [5] "1960.0"      "1961.0"      "1962.0"      "1963.0"
## [9] "1964.0"      "1965.0"      "1966.0"      "1967.0"
## [13] "1968.0"      "1969.0"      "1970.0"      "1971.0"
## [17] "1972.0"      "1973.0"      "1974.0"      "1975.0"
## [21] "1976.0"      "1977.0"      "1978.0"      "1979.0"
## [25] "1980.0"      "1981.0"      "1982.0"      "1983.0"
## [29] "1984.0"      "1985.0"      "1986.0"      "1987.0"
## [33] "1988.0"      "1989.0"      "1990.0"      "1991.0"
## [37] "1992.0"      "1993.0"      "1994.0"      "1995.0"
## [41] "1996.0"      "1997.0"      "1998.0"      "1999.0"
## [45] "2000.0"      "2001.0"      "2002.0"      "2003.0"
## [49] "2004.0"      "2005.0"      "2006.0"      "2007.0"
## [53] "2008.0"      "2009.0"      "2010.0"      "2011.0"
## [57] "2012.0"      "2013.0"      "2014.0"      "2015.0"
## [61] "2016.0"      "2017.0"      "2018.0"      "2019.0"
## [65] "2020.0"      "2021.0"      "2022.0"      "2023.0"
## [69] "2024.0"      "2025.0"
```

```
str(debt)
```

```
## tibble [197 x 217] (S3: tbl_df/tbl/data.frame)
## $ DEBT (% of GDP): chr [1:197] NA "Afghanistan" "Albania" "Algeria" ...
## $ 1800 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1801 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1802 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1803 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1804 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1805 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1806 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1807 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1808 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1809 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1810 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1811 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1812 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1813 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1814 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1815 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1816 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1817 : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1818 : chr [1:197] NA "no data" "no data" "no data" ...
```

[illegible]

```
## $ 1873      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1874      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1875      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1876      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1877      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1878      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1879      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1880      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1881      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1882      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1883      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1884      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1885      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1886      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1887      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1888      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1889      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1890      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1891      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1892      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1893      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1894      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1895      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1896      : chr [1:197] NA "no data" "no data" "no data" ...
## $ 1897      : chr [1:197] NA "no data" "no data" "no data" ...
## [list output truncated]
```

```
str(growth)
```

```
## tibble [265 x 70] (S3: tbl_df/tbl/data.frame)
## $ Country Name : chr [1:265] "Aruba" "Africa Eastern and Southern" "Afghanistan" "Africa Western and Central" ...
## $ Country Code : chr [1:265] "ABW" "AFE" "AFG" "AFW" ...
## $ Indicator Name: chr [1:265] "GDP growth (annual %)" "GDP growth (annual %)" "GDP growth (annual %)" "GDP growth (annual %)" ...
## $ Indicator Code: chr [1:265] "NY.GDP.MKTP.KD.ZG" "NY.GDP.MKTP.KD.ZG" "NY.GDP.MKTP.KD.ZG" "NY.GDP.MKTP.KD.ZG" ...
## $ 1960.0       : logi [1:265] NA NA NA NA NA NA ...
## $ 1961.0       : num [1:265] NA 0.419 NA 1.87 NA ...
## $ 1962.0       : num [1:265] NA 7.94 NA 3.73 NA ...
## $ 1963.0       : num [1:265] NA 5.62 NA 7.04 NA ...
## $ 1964.0       : num [1:265] NA 4.65 NA 5.36 NA ...
## $ 1965.0       : num [1:265] NA 5.14 NA 4.11 NA ...
## $ 1966.0       : num [1:265] NA 4.83 NA -1.51 NA ...
## $ 1967.0       : num [1:265] NA 5.34 NA -8.97 NA ...
## $ 1968.0       : num [1:265] NA 4.16 NA 1.57 NA ...
## $ 1969.0       : num [1:265] NA 5.12 NA 15.02 NA ...
## $ 1970.0       : num [1:265] NA 1.11 NA 17.31 NA ...
## $ 1971.0       : num [1:265] NA 5.48 NA 10.41 NA ...
## $ 1972.0       : num [1:265] NA 2.76 NA 3.17 NA ...
## $ 1973.0       : num [1:265] NA 4.62 NA 3.92 NA ...
## $ 1974.0       : num [1:265] NA 5.47 NA 9.92 NA ...
## $ 1975.0       : num [1:265] NA 1.37 NA -2.01 NA ...
## $ 1976.0       : num [1:265] NA 2.35 NA 8.42 NA ...
## $ 1977.0       : num [1:265] NA 1.11 NA 4.56 NA ...
## $ 1978.0       : num [1:265] NA 1.48 NA -1.92 NA ...
## $ 1979.0       : num [1:265] NA 2.93 NA 5.03 NA ...
```

```

## $ 1980.0      : num [1:265] NA 5.45 NA 1.87 NA ...
## $ 1981.0      : num [1:265] NA 3.93 NA -6.64 -4.4 ...
## $ 1982.0      : num [1:265] NA 0.309 NA -3.271 0 ...
## $ 1983.0      : num [1:265] NA 0.0633 NA -6.2893 4.2 ...
## $ 1984.0      : num [1:265] NA 3.406 NA 0.484 6 ...
## $ 1985.0      : num [1:265] NA -0.0863 NA 5.3199 3.5 ...
## $ 1986.0      : num [1:265] NA 2.27 NA 1.27 2.9 ...
## $ 1987.0      : num [1:265] 16.08 3.94 NA 1.41 4.08 ...
## $ 1988.0      : num [1:265] 18.65 4.29 NA 4.76 6.13 ...
## $ 1989.0      : num [1:265] 12.1298 2.6661 NA 1.7235 0.0416 ...
## $ 1990.0      : num [1:265] 3.961 0.121 NA 5.601 -3.45 ...
## $ 1991.0      : num [1:265] 7.963 -0.106 NA 1.118 0.991 ...
## $ 1992.0      : num [1:265] 5.88 -2.4 NA 2.29 -5.84 ...
## $ 1993.0      : num [1:265] 7.308 -0.839 NA -1.325 -23.983 ...
## $ 1994.0      : num [1:265] 8.204 1.922 NA -0.229 1.339 ...
## $ 1995.0      : num [1:265] 2.55 4.35 NA 1.85 15 ...
## $ 1996.0      : num [1:265] 1.19 5.5 NA 4.62 13.54 ...
## $ 1997.0      : num [1:265] 7.05 3.86 NA 4.39 7.27 ...
## $ 1998.0      : num [1:265] 1.99 1.77 NA 3.62 4.69 ...
## $ 1999.0      : num [1:265] 1.24 2.63 NA 1.52 2.18 ...
## $ 2000.0      : num [1:265] 7.62 3.18 NA 3.81 3.05 ...
## $ 2001.0      : num [1:265] 4.18 3.5 -9.43 5.25 4.21 ...
## $ 2002.0      : num [1:265] -0.945 3.917 28.6 9.943 13.666 ...
## $ 2003.0      : num [1:265] 1.11 3.01 8.83 5.63 3.49 ...
## $ 2004.0      : num [1:265] 7.29 5.63 1.41 8.07 11.42 ...
## $ 2005.0      : num [1:265] -0.383 6.174 11.23 5.797 14.155 ...
## $ 2006.0      : num [1:265] 1.13 6.64 5.36 5.29 11.84 ...
## $ 2007.0      : num [1:265] 3.09 6.7 13.83 5.45 13 ...
## $ 2008.0      : num [1:265] 1.84 4.44 3.92 6.21 10.79 ...
## $ 2009.0      : num [1:265] -11.678 0.919 21.391 6.141 1.996 ...
## $ 2010.0      : num [1:265] -2.73 5.34 14.36 7 5.29 ...
## $ 2011.0      : num [1:265] 3.369 3.791 0.426 4.922 3.594 ...
## $ 2012.0      : num [1:265] -1.04 1.88 12.75 5.13 8.5 ...
## $ 2013.0      : num [1:265] 6.43 4.63 5.6 6.05 4.88 ...
## $ 2014.0      : num [1:265] 1.43 3.92 2.72 5.69 4.66 ...
## $ 2015.0      : num [1:265] 3.611 2.934 1.451 2.933 0.763 ...
## $ 2016.0      : num [1:265] 1.234 2.293 2.26 0.176 -0.233 ...
## $ 2017.0      : num [1:265] 3.493 2.679 2.647 2.303 -0.165 ...
## $ 2018.0      : num [1:265] 3.212 2.706 1.189 2.9 -0.507 ...
## $ 2019.0      : num [1:265] 1.23 1.94 3.91 3.28 -1.06 ...
## $ 2020.0      : num [1:265] -23.94 -2.93 -2.35 -3.72 -5.27 ...
## $ 2021.0      : num [1:265] 14.83 4.45 -20.74 2.55 1.09 ...
## $ 2022.0      : num [1:265] 10.61 3.67 -6.24 4.48 3.56 ...
## $ 2023.0      : num [1:265] 9.52 1.94 2.27 3.64 1.32 ...
## $ 2024.0      : num [1:265] 6.81 2.79 1.87 4.6 4.95 ...
## $ 2025.0      : num [1:265] NA 3.74 NA 4.6 3.13 ...

```

```
colSums(is.na(debt))
```

```

## DEBT (% of GDP)      1800      1801      1802      1803
##                2          3          3          3
##            1804      1805      1806      1807      1808
##                3          3          3          3
##            1809      1810      1811      1812      1813

```

##	3	3	3	3	3
##	1814	1815	1816	1817	1818
##	3	3	3	3	3
##	1819	1820	1821	1822	1823
##	3	3	3	3	3
##	1824	1825	1826	1827	1828
##	3	3	3	3	3
##	1829	1830	1831	1832	1833
##	3	3	3	3	3
##	1834	1835	1836	1837	1838
##	3	3	3	3	3
##	1839	1840	1841	1842	1843
##	3	3	3	3	3
##	1844	1845	1846	1847	1848
##	3	3	3	3	3
##	1849	1850	1851	1852	1853
##	3	3	3	3	3
##	1854	1855	1856	1857	1858
##	3	3	3	3	3
##	1859	1860	1861	1862	1863
##	3	3	3	3	3
##	1864	1865	1866	1867	1868
##	3	3	3	3	3
##	1869	1870	1871	1872	1873
##	3	3	3	3	3
##	1874	1875	1876	1877	1878
##	3	3	3	3	3
##	1879	1880	1881	1882	1883
##	3	3	3	3	3
##	1884	1885	1886	1887	1888
##	3	3	3	3	3
##	1889	1890	1891	1892	1893
##	3	3	3	3	3
##	1894	1895	1896	1897	1898
##	3	3	3	3	3
##	1899	1900	1901	1902	1903
##	3	3	3	3	3
##	1904	1905	1906	1907	1908
##	3	3	3	3	3
##	1909	1910	1911	1912	1913
##	3	3	3	3	3
##	1914	1915	1916	1917	1918
##	3	3	3	3	3
##	1919	1920	1921	1922	1923
##	3	3	3	3	3
##	1924	1925	1926	1927	1928
##	3	3	3	3	3
##	1929	1930	1931	1932	1933
##	3	3	3	3	3
##	1934	1935	1936	1937	1938
##	3	3	3	3	3
##	1939	1940	1941	1942	1943
##	3	3	3	3	3
##	1944	1945	1946	1947	1948

##	3	3	3	3	3
##	1949	1950	1951	1952	1953
##	3	3	3	3	3
##	1954	1955	1956	1957	1958
##	3	3	3	3	3
##	1959	1960	1961	1962	1963
##	3	3	3	3	3
##	1964	1965	1966	1967	1968
##	3	3	3	3	3
##	1969	1970	1971	1972	1973
##	3	3	3	3	3
##	1974	1975	1976	1977	1978
##	3	3	3	3	3
##	1979	1980	1981	1982	1983
##	3	3	3	3	3
##	1984	1985	1986	1987	1988
##	3	3	3	3	3
##	1989	1990	1991	1992	1993
##	3	3	3	3	3
##	1994	1995	1996	1997	1998
##	3	3	3	3	3
##	1999	2000	2001	2002	2003
##	3	3	3	3	3
##	2004	2005	2006	2007	2008
##	3	3	3	3	3
##	2009	2010	2011	2012	2013
##	3	3	3	3	3
##	2014	2015			
##	3	3			

```
colSums(is.na(growth))
```

##	Country Name	Country Code	Indicator Name	Indicator Code	1960.0
##	0	0	0	0	265
##	1961.0	1962.0	1963.0	1964.0	1965.0
##	121	114	114	114	114
##	1966.0	1967.0	1968.0	1969.0	1970.0
##	111	107	105	105	105
##	1971.0	1972.0	1973.0	1974.0	1975.0
##	81	81	81	81	79
##	1976.0	1977.0	1978.0	1979.0	1980.0
##	76	75	70	70	69
##	1981.0	1982.0	1983.0	1984.0	1985.0
##	60	58	56	56	53
##	1986.0	1987.0	1988.0	1989.0	1990.0
##	52	50	46	45	43
##	1991.0	1992.0	1993.0	1994.0	1995.0
##	24	24	23	23	22
##	1996.0	1997.0	1998.0	1999.0	2000.0
##	21	21	19	19	19
##	2001.0	2002.0	2003.0	2004.0	2005.0
##	18	18	14	14	14
##	2006.0	2007.0	2008.0	2009.0	2010.0
##	14	13	12	9	8

```
##      2011.0      2012.0      2013.0      2014.0      2015.0
##      8      9      9      8      7
##      2016.0      2017.0      2018.0      2019.0      2020.0
##      8      8      7      8      8
##      2021.0      2022.0      2023.0      2024.0      2025.0
##      8      9      14      18      32
```

```
head(debt)
```

```
## # A tibble: 6 x 217
##   'DEBT (% of GDP)' '1800' '1801' '1802' '1803' '1804' '1805' '1806' '1807'
##   <chr>           <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 <NA>           <NA> <NA> <NA> <NA> <NA> <NA> <NA> <NA>
## 2 Afghanistan    no data no data no data no data no data no da~ no da~ no da~
## 3 Albania         no data no data no data no data no data no da~ no da~ no da~
## 4 Algeria         no data no data no data no data no data no da~ no da~ no da~
## 5 Angola          no data no data no data no data no data no da~ no da~ no da~
## 6 Anguilla        no data no data no data no data no data no da~ no da~ no da~
## # i 208 more variables: '1808' <chr>, '1809' <chr>, '1810' <chr>, '1811' <chr>,
## #   '1812' <chr>, '1813' <chr>, '1814' <chr>, '1815' <chr>, '1816' <chr>,
## #   '1817' <chr>, '1818' <chr>, '1819' <chr>, '1820' <chr>, '1821' <chr>,
## #   '1822' <chr>, '1823' <chr>, '1824' <chr>, '1825' <chr>, '1826' <chr>,
## #   '1827' <chr>, '1828' <chr>, '1829' <chr>, '1830' <chr>, '1831' <chr>,
## #   '1832' <chr>, '1833' <chr>, '1834' <chr>, '1835' <chr>, '1836' <chr>,
## #   '1837' <chr>, '1838' <chr>, '1839' <chr>, '1840' <chr>, '1841' <chr>, ...
```

```
head(growth)
```

```
## # A tibble: 6 x 70
##   'Country Name'      'Country Code' 'Indicator Name' 'Indicator Code' '1960.0'
##   <chr>             <chr>         <chr>           <chr>           <dbl>
## 1 Aruba             ABW           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 2 Africa Eastern and ~ AFE           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 3 Afghanistan       AFG           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 4 Africa Western and ~ AFW           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 5 Angola            AGO           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## 6 Albania           ALB           GDP growth (ann~ NY.GDP.MKTP.KD.~ NA
## # i 65 more variables: '1961.0' <dbl>, '1962.0' <dbl>, '1963.0' <dbl>,
## #   '1964.0' <dbl>, '1965.0' <dbl>, '1966.0' <dbl>, '1967.0' <dbl>,
## #   '1968.0' <dbl>, '1969.0' <dbl>, '1970.0' <dbl>, '1971.0' <dbl>,
## #   '1972.0' <dbl>, '1973.0' <dbl>, '1974.0' <dbl>, '1975.0' <dbl>,
## #   '1976.0' <dbl>, '1977.0' <dbl>, '1978.0' <dbl>, '1979.0' <dbl>,
## #   '1980.0' <dbl>, '1981.0' <dbl>, '1982.0' <dbl>, '1983.0' <dbl>,
## #   '1984.0' <dbl>, '1985.0' <dbl>, '1986.0' <dbl>, '1987.0' <dbl>, ...
```

We can see that there are some missing values and that we will have to clean this data. We can see the variable types, if there are any duplicate rows and column names as well

Step 4: Clean the data

```

clean_debt <- function(debt) {

  debt_clean <- debt |>
    filter(`DEBT (% of GDP)` == "Australia") |>
    select(`DEBT (% of GDP)`, `1960`:`1969`) |>
    pivot_longer(
      cols = `1960`:`1969`,
      names_to = "Year",
      values_to = "debt_pct_gdp"
    ) |>
    mutate(
      Country = `DEBT (% of GDP)`,
      Year = as.numeric(Year),
      debt_pct_gdp = na_if(debt_pct_gdp, "no data"),
      debt_pct_gdp = as.numeric(debt_pct_gdp)
    ) |>
    select(Country, Year, debt_pct_gdp)

  return(debt_clean)
}

```

```

clean_growth <- function(growth) {

  growth_clean <- growth |>
    filter(`Country Name` == "Australia") |>
    select(`Country Name`, `1960.0`:`1969.0`) |>
    pivot_longer(
      cols = `1960.0`:`1969.0`,
      names_to = "Year",
      values_to = "growth_pct_gdp"
    ) |>
    mutate(
      Country = `Country Name`,
      Year = as.numeric(Year),
      growth_pct_gdp = as.numeric(growth_pct_gdp)
    ) |>
    select(Country, Year, growth_pct_gdp)

  return(growth_clean)
}

```

```

debt_clean <- clean_debt(debt)
growth_clean <- clean_growth(growth)

```

```
debt_clean
```

```

## # A tibble: 10 x 3
##   Country    Year debt_pct_gdp
##   <chr>      <dbl>      <dbl>
## 1 Australia  1960          31.5
## 2 Australia  1961          30.3

```

```
## 3 Australia 1962      30.4
## 4 Australia 1963      29.3
## 5 Australia 1964      27.6
## 6 Australia 1965      NA
## 7 Australia 1966      41.2
## 8 Australia 1967      39.2
## 9 Australia 1968      38.2
## 10 Australia 1969     35.7
```

```
growth_clean
```

```
## # A tibble: 10 x 3
##   Country   Year growth_pct_gdp
##   <chr>    <dbl>      <dbl>
## 1 Australia 1960         NA
## 2 Australia 1961         2.48
## 3 Australia 1962         1.30
## 4 Australia 1963         6.21
## 5 Australia 1964         6.98
## 6 Australia 1965         5.98
## 7 Australia 1966         2.38
## 8 Australia 1967         6.31
## 9 Australia 1968         5.10
## 10 Australia 1969         7.05
```

Pre-processing

```
pre_process <- function(debt_clean, growth_clean) {

  combined <- debt_clean |>
    inner_join(
      growth_clean,
      by = c("Country", "Year")
    ) |>
    select(Country, Year, debt_pct_gdp, growth_pct_gdp)

  return(combined)
}
```

```
combined <- pre_process(debt_clean, growth_clean)
```

```
combined
```

```
## # A tibble: 10 x 4
##   Country   Year debt_pct_gdp growth_pct_gdp
##   <chr>    <dbl>      <dbl>      <dbl>
## 1 Australia 1960         31.5         NA
## 2 Australia 1961         30.3         2.48
## 3 Australia 1962         30.4         1.30
## 4 Australia 1963         29.3         6.21
## 5 Australia 1964         27.6         6.98
## 6 Australia 1965         NA         5.98
```

##	7	Australia	1966	41.2	2.38
##	8	Australia	1967	39.2	6.31
##	9	Australia	1968	38.2	5.10
##	10	Australia	1969	35.7	7.05